

# Stress-based topology optimization for continua

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Received: 2 June 2009 / Revised: 20 August 2009 / Accepted: 5 September 2009 / Published online: 6 October 2009  
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**Abstract** We propose an effective algorithm to resolve the stress-constrained topology optimization problem. Our procedure combines a density filter for length scale control, the solid isotropic material with penalization (SIMP) to generate black-and-white designs, a SIMP-motivated stress definition to resolve the stress singularity phenomenon, and a global/regional stress measure combined with an adaptive normalization scheme to control the local stress level.

**Keywords** Topology optimization · Stress constraints · Regional stress measure · Normalization

## 1 Introduction

Topology optimization using the material distribution technique is a well established research field. However, most of the development has concentrated on compliance and other global response constraints such as frequency despite

the fact that stress is among the most important considerations. Most likely, this is because there are three significant challenges that need to be overcome to effectively solve stress-based optimization problems (Bendsøe and Sigmund 2003). The first is related to the so-called “singularity” phenomenon (Rozvany 2001; Cheng and Guo 1997; Cheng and Jiang 1992), the second is related to the local nature of the constraint, and the third is related to the highly non-linear stress behavior.

The “singularity” problem was first encountered when designing trusses subject to stress constraints where it was shown that the  $n$ -dimensional feasible design space contains degenerate subspaces of dimension less than  $n$  (Cheng and Jiang 1992; Kirsch 1990). Further the globally optimal design is often an element of such degenerate subspaces. Nonlinear programming algorithms cannot identify these regions and hence they converge to locally optimal designs. To remedy this situation, the stress constraints are relaxed to eliminate the degenerate regions and thereby allow the nonlinear programming to find the global optimum. Several relaxation approaches have been forwarded for truss design, e.g. the  $\varepsilon$ -relaxation and smooth envelope functions (SEF's) (Cheng and Guo 1997; Rozvany and Sobieszczanski-Sobieski 1992). These approaches were later adapted for the stress constrained design of continuum structures (Duysinx and Bendsøe 1998; Bruggi 2008; Duysinx and Sigmund 1998).

The second difficulty of stress-based topology optimization is due to the local nature of the stress constraint. In a continuum setting, stress constraints should be considered at every material point. Albeit in a discrete setting, the number of such material points is finite, it is still too large for practical applications. A resolution to this complication replaces the local stress constraints with a single integrated stress constraint that approximates the maximum stress, such as

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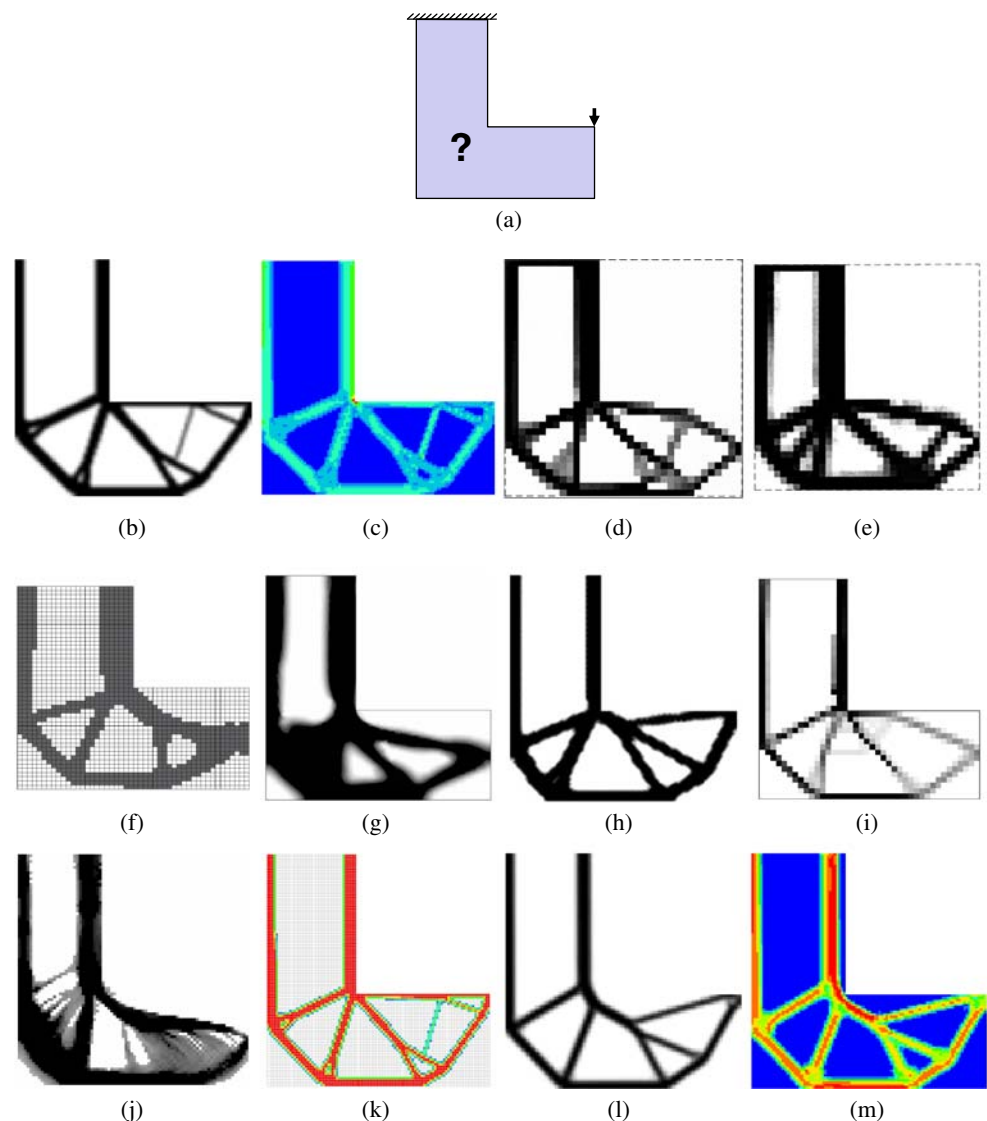
the  $p$ -norm and the *Kresselmeier–Steinhauser* (KS) functions (Duysinx and Sigmund 1998; Yang and Chen 1996). This global approach is computationally efficient, however it does not adequately control the local stress behavior.

Another issue with stress constraints is their highly nonlinear dependence on the design. Indeed, stress levels are drastically affected by density changes in neighboring regions, and this phenomenon is exasperated in critical regions with large spatial stress gradients such as reentrant corners. As such, design optimization problem formulations and their solution algorithms should be numerically consistent to avoid convergence problems.

Our goal is to design black-and-white structures, i.e. practical-inexpensive structures with homogeneous isotropic material properties, that satisfy stress constraints. Hence, we consider neither composite nor functionally graded materials (Lipton 2002; Stump et al. 2007; Allaire

et al. 2004). As a prelude to and motivation for our method, we present L-bracket stress-based topology designs for structures comprised of homogeneous isotropic materials, cf. Fig. 1. Table 1 summarizes the critical aspects that are used to obtain each design, namely the length scale control, material model used in the finite element analysis, relaxed stress used to define the stress constraint, problem statement, nonlinear programming algorithm as well as any extraneous remarks. In all cases the L-bracket is fixed at the top edge, and subjected to a vertical transverse tip load. In some designs, the load is applied at the top corner as shown, while in others it is applied at the center of the rightmost vertical edge. The initial design consists of a uniform material distribution, and hence it includes the reentrant corner with its stress singularity. For reference purposes, a minimum compliance design (subject to a volume constraint) is also provided along with its stress contour plot, cf. design

**Fig. 1** Various L-bracket designs: **b** and **c** are compliance based; **d–m** are stress-based. (Stress plots are based on relaxed von Mises stress, cf. Section 3, and have different color scale ranges). **a** Problem definition, **b** Min. compliance design, **c** Stress distribution in **b**, **d** Duysinx and Bendsøe (1998), **e** Duysinx and Sigmund (1998), **f** Svanberg and Werme (2007), **g** Pereira et al. (2004), **h** Bruggi and Venini (2008), **i** Paris et al. (2009), **j** Guilherme and Fonseca (2007), **k** Altair Optistruct (2007) (density distribution), **l** Present work, **m** Stress distribution in **l**



**Table 1** Algorithms used to obtain the designs in Fig. 1 (the solid stress is  $\mathbf{T}_o = \mathbb{C}_o [\nabla u]$  where  $\mathbb{C}_o$  is the solid elasticity tensor and  $u$  is the displacement)

| Author/design                             | Length scale control                                                                            | Elasticity tensor                                               | Relaxed stress                                                                                                                                           | Problem statement                                                                    | Optimization algorithm                              | Remarks                                                                                                                 |
|-------------------------------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Duysinx and Bendsøe (1998) (d)            | Not evident                                                                                     | SIMP (Bendsøe 1989)<br>$\mathbb{C}(\rho) = \rho^p \mathbb{C}_o$ | $\varepsilon$ -relaxation; variant of Cheng and Guo (1997)<br>$\mathbf{T}_r = \frac{\rho}{\varepsilon + \rho} \mathbf{T}_o$                              | Min. volume s.t. local stress constraints                                            | CONLIN (Fleury and Braibant 1986)                   | Models porous material with checkerboard control via Diaz and Sigmund (2007)                                            |
| Duysinx and Sigmund (1998) (e)            | Sensitivity filter (Sigmund 1997)                                                               | SIMP (Bendsøe 1989)<br>$\mathbb{C}(\rho) = \rho^p \mathbb{C}_o$ | $\varepsilon$ -relaxation; variant of Cheng and Guo (1997)<br>$\mathbf{T}_r = \frac{\rho}{\varepsilon(1-\rho) + \rho} \mathbf{T}_o$                      | Min. volume s.t. global stress constraints                                           | MMA (Svanberg 1987)                                 |                                                                                                                         |
| Svanberg and Werme (2007) (f)             | Not evident                                                                                     | $\mathbb{C}(\rho) = \rho \mathbb{C}_o$                          | $\mathbf{T}_r = \rho \mathbf{T}_o$                                                                                                                       | Min. volume weighted by the max stress and min. volume s.t. local stress constraints | Sequential linear and quadratic integer programming | Solves discrete $\{0, 1\}$ problem; Second order elements and “linear inequality” constraints for checker-board control |
| Pereira et al. (2004) (g)                 | Perimeter control (Haber et al. 1996)/regularized density control (Borrvall and Petersson 2001) | SIMP (Bendsøe 1989)<br>$\mathbb{C}(\rho) = \rho^p \mathbb{C}_o$ | $\varepsilon$ -relaxation; variant of Cheng and Guo (1997)<br>$\mathbf{T}_r = \frac{\rho}{\varepsilon(1-\rho) + \rho} \mathbf{T}_o$                      | Min. volume s.t. local stress constraints                                            | BOX-QUACAN (Friedlander et al. 1994)                | Augmented Lagrangian formulation                                                                                        |
| Bruggi and Venini (2008) (h)              | Not evident                                                                                     | SIMP (Bendsøe 1989)<br>$\mathbb{C}(\rho) = \rho^p \mathbb{C}_o$ | $qp$ relaxation; variant of $\varepsilon$ -relaxation; requires continuation method (Bruggi 2008)<br>$\mathbf{T}_r = \frac{\rho^p}{\rho^q} \mathbf{T}_o$ | Min. compliance s.t. volume and local stress constraints                             | MMA (Svanberg 1987)                                 | Mixed finite elements; nodal density variables for checker-board control (Rahmatalla and Swan 2004)                     |
| Paris et al. (2009) (i)                   | Not evident                                                                                     | $\mathbb{C}(\rho) = \rho \mathbb{C}_o$                          | $\varepsilon$ -relaxation; variant of Cheng and Guo (1997)<br>$\mathbf{T}_r = \frac{\rho}{\varepsilon(1-\rho) + \rho} \mathbf{T}_o$                      | Min. volume s.t. local stress constraints                                            | Sequential linear programming                       | Volume SIMP approach (Zhou and Rozvany 1991)<br>$V = \int_{\Omega} \rho^p dv$<br>$0 < p < 1$                            |
| Guilherme and Fonseca (2007) (j)          | Filter (Fonseca and Cardoso 2003)                                                               | SIMP (Bendsøe 1989)<br>$\mathbb{C}(\rho) = \rho^p \mathbb{C}_o$ | Alternative $\varepsilon$ -relaxation (Guilherme and Fonseca 2007)<br>$\mathbf{T}_r = \frac{1}{1 + \varepsilon^p - \varepsilon} \mathbf{T}_o$            | Min. volume s.t. global stress constraints                                           | Sequential linear programming                       |                                                                                                                         |
| Optipstruct (Altair Engineering 2007) (k) | Filter (Zhou et al. 2001)                                                                       | Not available                                                   | Not available                                                                                                                                            | Min. volume s.t. global stress constraints                                           | Not available                                       | Proprietary software                                                                                                    |
| Present work (l)                          | Density filter (Bruns and Tortorelli 2001)                                                      | $\mathbb{C}(\rho) = \eta_c(\rho) \mathbb{C}_o$                  | $\mathbf{T}_r = \eta_T(\rho) \mathbf{T}_o$                                                                                                               | Min. volume s.t. regional-normalized stress constraints                              | MMA (Svanberg 1987)                                 | Also, min. stress s.t. volume constraint                                                                                |

(b) and plot (c). All stress plots in this paper use color scales that range from blue (minimum stress) to red (maximum stress), with nine equal range levels, and unless noted, all stress plots in any one figure use the same color scale. Note that the reentrant corner is not removed in the compliance design. As seen in Fig. 1, designs (d), (e), (h), (i) and (k) have reentrant corners, and in fact they are quite similar to the compliance design (b). The reentrant corner is removed in the designs (f), (g), (j) and our design (l) (with accompanying stress distribution plot (m)). Unfortunately, design (f) is obtained via an integer programming method which is not yet viable for large scale applications. Designs (f) and (h) do not contain length scale control (albeit they contain checkerboard control) and hence their problem formulations are most likely ill-posed, and design (j) contains large regions of gray material. The approach in design (g) is promising, but it requires the tuning of parameters which can heavily influence the final topology, and the authors report that their stress-based problem is significantly more computationally expensive than the compliance problem.

In this work, we propose a practical procedure to solve the stress-constrained topology optimization (Le et al. 2009), it combines (1) a density filter for length scale control, (2) a solid isotropic material with penalization (SIMP) model (Bendsøe 1989; Zhou and Rozvany 1991; Rozvany et al. 1992) to generate black-and-white designs, (3) a SIMP-like stress definition to resolve the stress singularity phenomenon, (4) global and regional stress measures and (5) an adaptive normalization scheme to precisely control the local stress level. The first two ingredients are well known in topology optimization (Bruns and Tortorelli 2001) while the third ingredient is consistent with stress-based topology optimization methodologies (Duysinx and Bendsøe 1998; Bruggi and Venini 2008). The combination of multiple regional stress measures and adaptive scaling yields designs that precisely satisfy the stress limits.

The remainder of the paper is organized as follows. Sections 2 and 3 discuss the density filter and the SIMP-like stress definition. The  $p$ -norm global stress measure is reviewed in Section 4. Sections 5 and 6 describe our normalization procedure and regional stress measure. In Section 7, we show how the normalized global/regional measures can be adapted to solve problems with multiple local constraints and multiple load cases. A summary of our algorithm and additional examples are provided in Section 8. Finally, conclusions are drawn in Section 9.

## 2 Density filtering

The density filtering technique was introduced by Bruns and Tortorelli (2001) and later proven by Bourdin (2001) to generate a well-posed topology optimization problem. In this

method the design field  $d$  is filtered to define the material density field  $\rho$ . With our piecewise uniform finite element discretization of density, the vectors  $\mathbf{d}$  and  $\mathbf{\rho}$  contain the element design variables and densities, respectively. The latter are defined by filtering the former in some way. As one example, the filter can be defined by:

$$\rho_i = \frac{\sum_{j \in \Omega_i} w_j d_j}{\sum_{j \in \Omega_i} w_j} \quad \text{with} \quad w_j = \frac{r_o - r_j}{r_o} \quad (1)$$

where the domain  $\Omega_i$  of element  $i$  contains all elements  $j$  that lie within the radius  $r_o$  of element  $i$  as measured from their respective centroids. The weighting factor  $w_j > 0$  as defined above is that of a cone filter where  $r_j$  is the distance between the element  $i$  and  $j$  centroids. Alternative smoothing filters may work as well.

In topology optimization, the density  $\rho$  is bounded between 0 and 1. We impose the same 0 and 1 bounds on  $d$  and hence by defining  $\rho$  via the filter, we achieve an upper bound value for  $|\nabla \rho|$  where  $\nabla \rho$  is the spatial gradient of  $\rho$ . In other words, by defining  $\rho$  through the bounded  $d$  we impose smoothness on  $\rho$  without requiring any additional constraints on  $d$ . This smoothness prohibits designs with small scale features such as narrow members, jagged edges, micro perforations, and sharp interfaces. Without this or a similar restriction on  $\rho$  the topology optimization problem is ill-posed (Bendsøe and Sigmund 2003).

Much effort in density-based topology optimization is directed towards the generation of black-and-white designs from which structural members can be readily identified (Bendsøe and Sigmund 2003; Svanberg and Werme 2007; Bruns 2005; Sigmund 2007). However, the generation of black-and-white designs that exhibit jagged edges at early, intermediate iterations may not be preferable for stress-based topology optimization. Indeed, stress is calculated from the displacement gradient which is computed with less accuracy than the finite element displacement field, especially in stress concentration regions. Therefore, the computed stress at element centroids is artificially low in the jagged regions which are known to contain stress singularities. Ultimately, these jagged boundaries are smoothed in the design interpretation stage whence the computed stress levels increase. Thus, it may be better to generate designs with smooth boundaries from the onset, and hence a somewhat blurred smooth boundary generated with the filter of (1) is preferable over a sharp jagged boundary.

The filter-induced blurred region around structural members in topology designs has another advantage: it facilitates the evolution of the design towards the global optima by allowing void elements to become solid elements more easily and vice-versa. It is the SIMP method (Bendsøe 1989; Zhou and Rozvany 1991; Rozvany et al. 1992)

which penalizes intermediate densities to generate black-and-white designs and can make the optimization prematurely converge to local optimal (Sigmund and Petersson 1998). In this sense, the filter provides smoothing for the optimization algorithm which enhances its ability to converge to the global optima. And for this reason, the filter and SIMP methods are often combined (Bendsøe and Sigmund 2003; Bruns and Tortorelli 2001).

As mentioned above, stress constraints are highly non-linear in the design. For this reason we implement the consistent filter introduced in Bruns and Tortorelli (2001). In this way, the constraint sensitivities are numerically consistent with the optimization problem statement and thereby the convergence behavior of the nonlinear programming algorithm is not unduly sacrificed. We have also implemented the popular and somewhat heuristic sensitivity filter introduced in Sigmund (1997) and found that, while it performs well for compliance problems, it does not perform particularly well for stress-constrained problems.

### 3 Relaxed stress

The “singularity” problem and its treatment have been extensively studied for stress-constrained truss problems (Rozvany 2001; Cheng and Guo 1997; Cheng and Jiang 1992; Kirsch 1990; Rozvany and Birker 1994). The problem manifests itself when the optimal topologies belong to degenerate subspaces of the feasible design space where one or more bars have zero cross section; and hence these optimal topologies are referred to as singular topologies (Cheng and Jiang 1992; Kirsch 1990; Rozvany and Birker 1994). Unfortunately, convergence to these singular topologies is practically impossible with gradient based optimizers. To eliminate the degenerate subspaces, the stress constraints are relaxed, cf. the  $\varepsilon$ -relaxation and smooth envelope functions (SEF's) for truss design (Cheng and Guo 1997; Rozvany and Sobieszczanski-Sobieski 1992), and their variations for continuum design (Duysinx and Bendsøe 1998; Bruggi 2008; Duysinx and Sigmund 1998). The motivation for introducing the relaxation is purely to make the topology optimization problem more tractable by generating a smooth feasible design space; this is the same reason that intermediate densities  $\rho \in [0, 1]$  are used in place of the discrete density  $\rho \in \{0, 1\}$ .

Here, our goal is to design black-and-white structures with homogeneous isotropic material properties that satisfy stress constraints; this is the reason that we adopt the SIMP method. We are not particularly concerned with the fact that the SIMP material exists for intermediate density values because ultimately we consider only black-and-white designs. Nonetheless, we note that the SIMP can be used to

model porous microstructures indicative of materials with intermediate density values (Bendsøe and Sigmund 1999).

Continuing with our black-and-white design goal, we now introduce our SIMP-like relaxed stress definition. To a degree it is consistent with the  $\varepsilon$ -relaxation used in Duysinx and Bendsøe (1998). However, in Duysinx and Bendsøe (1998) much concern is directed towards generating a physically consistent scheme for intermediate density values; perhaps not surprisingly, their relaxed stress measure corresponds to the stress in a porous microstructure. We make no attempt to accurately represent the behavior of materials with intermediate density values; again because our goal is to consider only black-and-white designs.

Our approach is simple. To begin with, the discrete topology optimization problem, i.e. where  $\rho \in \{0, 1\}$ , is relaxed to a continuous optimization problem, i.e. where  $\rho \in [0, 1]$ , which is forced to generate a discrete design at the end of the optimization process. Consequently, we need to define interpolation schemes for the stiffness, volume, and stress to interpolate between their possible minimum and maximum values. For the stiffness, we introduce  $\eta_c$  to weight the solid material elasticity tensor  $\mathbb{C}_o$ , i.e.

$$\mathbb{C}(\rho) = \eta_c(\rho) \mathbb{C}_o \quad (2)$$

such that  $\eta_c$  satisfies (1)  $\eta_c$  is a monotonically increasing function, (2)  $0 < \eta_c(\rho) \leq \rho$  for  $0 < \rho < 1$ , (3)  $\eta_c(1) = 1$  and (4)  $\eta_c(0) = 0$ . For the volume, we introduce  $\eta_v$  to weight the infinitesimal volume  $dv_o$  as:

$$dv = \eta_v(\rho) dv_o \quad (3)$$

where the total volume is computed as  $V = \int_{\Omega} \eta_v(\rho) dv_o$  and such that  $\eta_v$  satisfies (1)  $\eta_v$  is a monotonically increasing function, (2)  $0 \leq \eta_v(\rho) < 1$  for  $0 < \rho < 1$ , (3)  $\eta_v(1) = 1$  and (4)  $\eta_v(0) = 0$ . We note that the interpolation of stiffness and volume are mainly responsible for the penalization of intermediate densities to force the final design to discrete. For example, the interpolations corresponding to the SIMP method are  $\eta_c(\rho) = \rho^\rho$  and  $\eta_v(\rho) = \rho$  whereby the stiffness is penalized for intermediate densities. Alternatively, we may penalize the volume, or both the stiffness and volume, as in the “sinh” method (Bruns 2005).

Similarly, we introduce  $\eta_T$  to weight the stress in the solid material  $\mathbf{T}_o \equiv \mathbb{C}_o [\nabla u]$ , i.e.

$$\mathbf{T}_r(\rho) = \eta_T(\rho) \mathbf{T}_o \quad (4)$$

such that (1)  $\eta_T$  is a monotonically increasing function, (2)  $\eta_c(\rho) < \eta_T(\rho) < 1$  for  $0 < \rho < 1$ ; in our work we define  $\rho < \eta_T(\rho)$  so that intermediate densities are further penalized by the stress interpolation, (3)  $\eta_T(0) = 0$  so that (a) the stress in void regions is zero and (b) the feasible design space is smooth without degenerate regions and

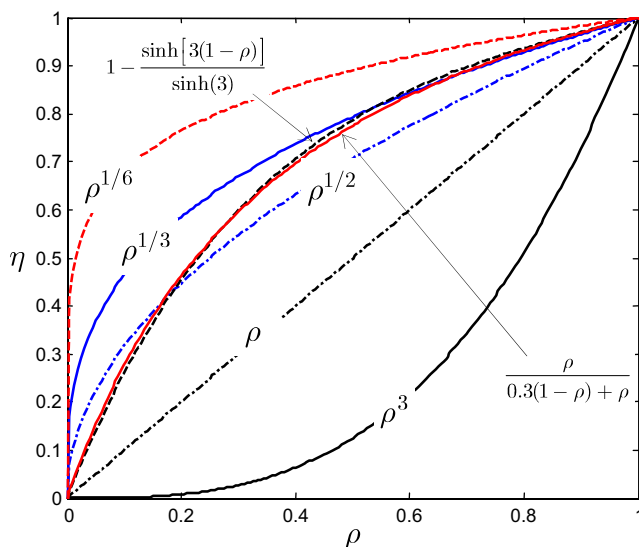


Fig. 2 Interpolation functions in (5)

(4)  $\eta_T(1) = 1$  so that the relaxed stress is consistent with stress in the solid material.

The bounds on  $\eta_T$  in condition 2 for (4) imply that our interpolated stress is chosen between two limiting stresses: (a) the solid stress defined as  $\mathbf{T}_o \equiv \mathbb{C}_o[\nabla \mathbf{u}]$ , and (b) the macroscopic stress defined as  $\mathbf{T}(\rho) \equiv \mathbb{C}(\rho)[\nabla \mathbf{u}] = \eta_c(\rho) \mathbf{T}_o$ . Both of these stresses are not suitable for use in stress-based topology optimization. Considering the stress in case (a), we have non-zero stress at zero densities since the strain  $\nabla \mathbf{u}$  is usually non-zero, and hence the optimizer will be unable to eliminate material in some areas of the design domain. Considering the stress in case (b) will lead to the opposite problem: the optimizer will generate a trivial all-void design. To see this, we consider a feasible design  $\rho$  which has the elasticity tensor  $\mathbb{C}(\rho) = \eta_c(\rho) \mathbb{C}_o$  where we assume that  $\eta_c(\rho)$  is a homogeneous function of degree  $p > 1$ , e.g.  $\eta_c(\rho) = \rho^p$ , and we have the displacement  $u(\rho)$ . Now, we suppose the feasible design  $\rho$  is uniformly scaled such that  $\rho \rightarrow \alpha\rho$  where  $0 < \alpha < 1$ . For this scaled design, we have  $\mathbb{C}(\alpha\rho) = \eta_c(\alpha\rho) \mathbb{C}_o = \alpha^p \eta_c(\rho) \mathbb{C}_o = \alpha^p \mathbb{C}(\rho)$  and hence  $u(\alpha\rho) = \alpha^{-p} u(\rho)$ . We see that considering the stress in case (b) gives  $\mathbf{T}(\alpha\rho) = \mathbf{T}(\rho)$ . If we try to uniformly eliminate material in case (b), the stress is unaffected, and hence the optimizer will attempt to remove all material.

Some remarks are in order: (1) the relaxed stress  $\mathbf{T}_r$  is used to define the stress measure  $\sigma$ , e.g. the von Mises stress, that in turn can be used to define either the stress constraints or the objective function, (2) the ultimate  $\eta_T$  enforced in Duysinx and Bendsøe (1998) and Bruggi and Venini (2008) via continuation leads to functions that approximate the step function, with nearly zero sensitivity

(for a fixed strain) which may adversely affect the optimization convergence, (3) the choices of  $\eta_c$ ,  $\eta_v$ , and  $\eta_T$  are flexible and (4) we have experimented with the following interpolation schemes (based on Cheng and Guo 1997; Bruggi 2008; Bendsøe 1989; Sigmund 1997; Zhou and Rozvany 1991) that work for the L-bracket example:

$$\begin{aligned}
 &\eta_c = \rho^3 \quad \eta_c = \rho \quad \eta_c = \rho^3 \\
 &1) \eta_v = \rho \quad , \quad 2) \eta_v = \rho^{1/3}, \quad 3) \eta_v = \rho \\
 &\eta_T = \rho^{1/2} \quad \eta_T = \rho^{1/6} \quad \eta_T = 1 - \frac{\sinh[3(1-\rho)]}{\sinh(3)} \\
 &\text{and} \quad \eta_c = \rho^3 \\
 &4) \eta_v = \rho \\
 &\eta_T = \frac{\rho}{0.3(1-\rho) + \rho} \quad (5)
 \end{aligned}$$

Figure 2 shows graphs of the functions of the four interpolation schemes in (5). We used the first interpolation scheme to generate the results presented in this paper.

#### 4 Global stress measure

In finite element-based stress-constrained topology optimization problems we ideally enforce one stress constraint per finite element. This implies the number of stress constraints  $n$  is large. Further, since the number of design variables also equals the number of elements, the number of design variables  $n$  is also large. Hence sensitivity computation, by either the direct or adjoint methods is prohibitively costly. Indeed, the number of pseudo analyses in the direct method and the number of adjoint analyses in the adjoint method equal  $n$  (Michaleris et al. 1994). Thus, both methods are equally costly. To reduce this computational burden a single global stress measure (Duysinx and Sigmund 1998; Yang and Chen 1996) is often used in place of the  $n$  local stress measures thereby making the adjoint sensitivity analysis computationally efficient.

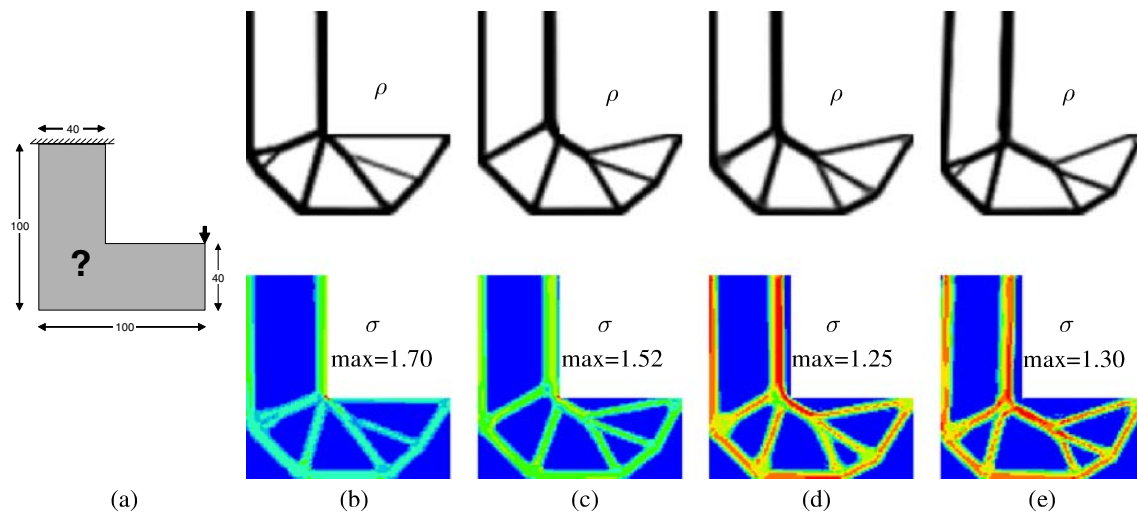
To define the global measure we first consider the original problem with the  $n$  constraints, one for each element  $e$ :

$$\sigma_e \leq \bar{\sigma}, \quad e = 1, 2, \dots, n \quad (6)$$

where  $\sigma \equiv \hat{\sigma}(\mathbf{T}_r)$  is the relaxed stress measure, e.g. von Mises stress,  $\bar{\sigma}$  is its bound, and  $e$  is the element index. These  $n$  constraints can be restated in terms of the single maximum stress constraint:

$$\sigma_{\max}(\rho) = \max_{e=1..n} (\sigma_e(\rho)) \leq \bar{\sigma} \quad (7)$$

However, the *maximum* function is not differentiable, thus it needs to be smoothed. For this purpose, one could use the  $p$ -norm or the Kreisselmeier–Steinhauser (KS) function.



**Fig. 3** Effect of stress norm parameter  $P$  on the L-bracket. **a** Problem definition, **b**  $P = 4$  (63 iter.), **c**  $P = 6$  (72 iter.), **d**  $P = 8$  (78 iter.), **e**  $P = 12$  (235 iter.)

Without loss of generality we adopt the  $p$ -norm measure  $\sigma_{PN}$ , and hence our constraint becomes:

$$\sigma_{PN} = \left( \sum_{e=1}^N v_e \sigma_e^P \right)^{1/P} \leq \bar{\sigma}_{PN} \quad (8)$$

where  $P$  is the stress norm parameter and  $v_e$  is the element  $e$  solid volume. Here we assume that a single stress criterion is imposed in the structure and that the stress measure  $\sigma_e$  is non-negative. For more generality, refer to Section 7.

On one extreme, as the stress norm parameter  $P \rightarrow \infty$ , the  $p$ -norm  $\sigma_{PN}$  approaches the maximum stress  $\sigma_{\max}$  (modulo the element volume), and there is no added smoothness. On the other extreme, when  $P \rightarrow 1$ , there is excessive smoothness but the  $p$ -norm approaches the average stress (modulo the volume). A good choice for  $P$  should therefore provide adequate smoothness so that the optimization algorithm performs well and an adequate approximation of the maximum stress value so that the optimized design satisfies the imposed stress constraints.

For stress minimization formulations the choice of the stress norm parameter  $P$  is not critical because the  $p$ -norm only needs to capture the trend of the maximum stress, not the actual maximum stress value. To see this we now design the L-bracket described in Fig. 3a. Unless noted, all dimensions given in this paper are in mm. Our problem is formulated to minimize stress  $\sigma_{PN}$ , cf. (8), where  $\sigma_e$  is the von Mises stress at the element centroids (elemental von Mises stresses), subject to the volume constraint  $V \leq \bar{V} \leq V_{\max}$  where  $V_{\max}$  is the design domain volume; it is solved via the method of moving asymptotes (MMA) (Svanberg 1987). The plane stress finite element analysis

uses bilinear 4-node square elements with a thickness of 1.0 mm and edge length of 1.0 mm; the material has a Young's modulus of  $E = 1.0$  MPa and a Poisson's ratio of  $\nu = 0.3$ ; the cone filter radius is  $r_i = 2.0$  mm, cf. (1); and the 3 N load is distributed over six nodes to avoid a stress concentration. Figure 3b–e show designs obtained using the stress norm parameter values  $P = 4, 6, 8$ , and 12 together with stress distribution plots. Unless noted, all stress values given in this paper are in MPa. As seen, a low  $P$  value gives a design similar to that of the compliance minimization, which has a stress concentration at the reentrant corner. On the other hand, larger  $P$  values give designs with more uniform stress distributions in which the stress concentration is reduced. We note however that the optimization with  $P = 12$  takes more iterations, and is probably more susceptible to local minima, in comparison to the optimization with lower  $P$  values. For this stress minimization example the values of  $P = 6$  and 8 yield best designs. We also note that the number of iterations required to obtain these designs is similar to that required for the compliance problem.

## 5 Normalized global stress measure

Unfortunately the  $p$ -norm stress measure  $\sigma_{PN}$  lacks physical meaning as opposed to, e.g. the maximum stress  $\sigma_{\max}$ . In fact, an explicit mathematical expression of  $\sigma_{\max}$  in terms of  $\sigma_{PN}$  does not exist (Duysinx and Sigmund 1998). Thus the ability to enforce a constraint on the maximum stress using the  $p$ -norm stress  $\sigma_{PN}$  is lacking. To remedy this deficiency we propose a normalized global stress measure to better approximate the maximum stress.

Our normalized global stress measure uses information from the previous optimization iteration to scale, i.e. normalize, the global  $p$ -norm measure as  $c \sigma_{PN}$  so that it better approximates the maximum stress. To define the normalization parameter  $c$  we use the maximum stress  $\sigma_{\max}^{I-1}$  and the  $p$ -norm  $\sigma_{PN}^{I-1}$  values from the previous optimization iteration  $I-1$  and define our evolving normalized global  $p$ -norm constraint at each iteration  $I$  as:

$$\sigma_{\max} \approx c \sigma_{PN} \leq \bar{\sigma} \quad (9)$$

where  $c$  is calculated at each optimization iteration  $I \geq 1$  as follows:

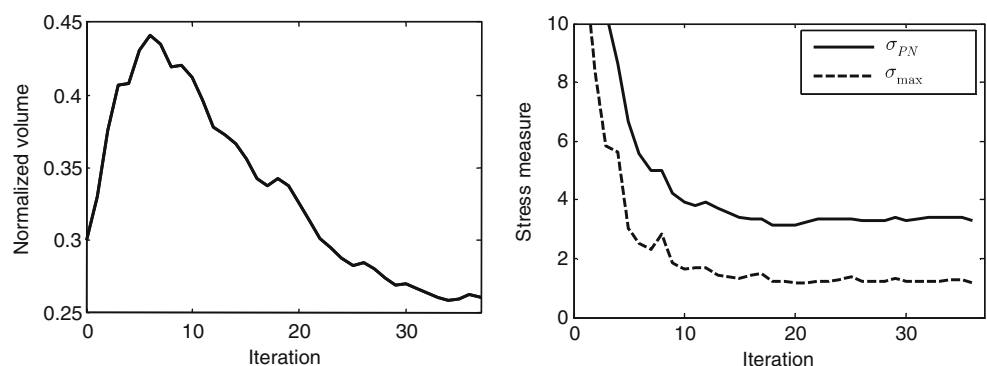
$$c^I = \alpha^I \frac{\sigma_{\max}^{I-1}}{\sigma_{PN}^{I-1}} + (1 - \alpha^I) c^{I-1} \quad (10)$$

The above parameter  $\alpha^I \in (0, 1]$  controls the variations between  $c^I$  and  $c^{I-1}$ :  $0 < \alpha^I < 1$  is chosen if  $c$  tends to oscillate between iterations, otherwise  $\alpha^I = 1$  is chosen. Obviously, we select  $\alpha^1 = 1$ . As the design converges,  $\mathbf{d}^I \approx \mathbf{d}^{I-1}$  so that  $\sigma_{PN}^I \approx \sigma_{PN}^{I-1}$  and  $\sigma_{\max}^I \approx \sigma_{\max}^{I-1}$  and hence we achieve the desired effect, i.e.  $c^I \sigma_{PN}^I \approx \sigma_{\max}^I$ .

Note that this constraint  $c \sigma_{PN} \leq \bar{\sigma}$  is non-differentiable because the value of  $c$  is changed in a discontinuous manner and results in a slightly different optimization problem at every iteration. However, as the optimization converges the changes between successive design iterations diminish and hence  $c$  converges, thereby reducing the effects of the non-differentiability and inconsistency. We observed in all of our computations that they do not cause convergence issues.

To demonstrate the normalization scheme, we revisit the L-bracket example of Fig. 3. Here, however, we minimize volume subject to a maximum elemental von Mises stress of 1.2 MPa, i.e.  $c \sigma_{PN} \leq \bar{\sigma} = 1.2$  where the stress norm parameter is  $P = 6$ . Figures 4, 5 and 6 show the design density and stress distribution evolutions together with the volume and stress convergence. From the convergence graphs it is seen that the maximum stress constraint is active in the optimal design.

**Fig. 4** Normalized volume (left) and stress (right) convergence



## 6 Regional stress measures

Thus far we have discussed the use of many local or one global stress measure(s) for stress-constrained problems. Imposing local stress constraints provides precise control over local stress levels, but it requires a prohibitively large computational expense. Alternatively, using one global stress measure relieves the computational expense, but it provides poor local control over the stress distribution. Our compromise between the two approaches uses several regional stress measures to improve the local control while maintaining manageable computational cost. So rather than using the  $n$  constraints of (6) or the single global constraint of (9), we enforce the  $m$  regional constraints

$$\sigma_{\max_k} = \max_{e \in \Omega_k} (\sigma_e) \leq \bar{\sigma}, \quad k = 1, 2, \dots, m \quad (11)$$

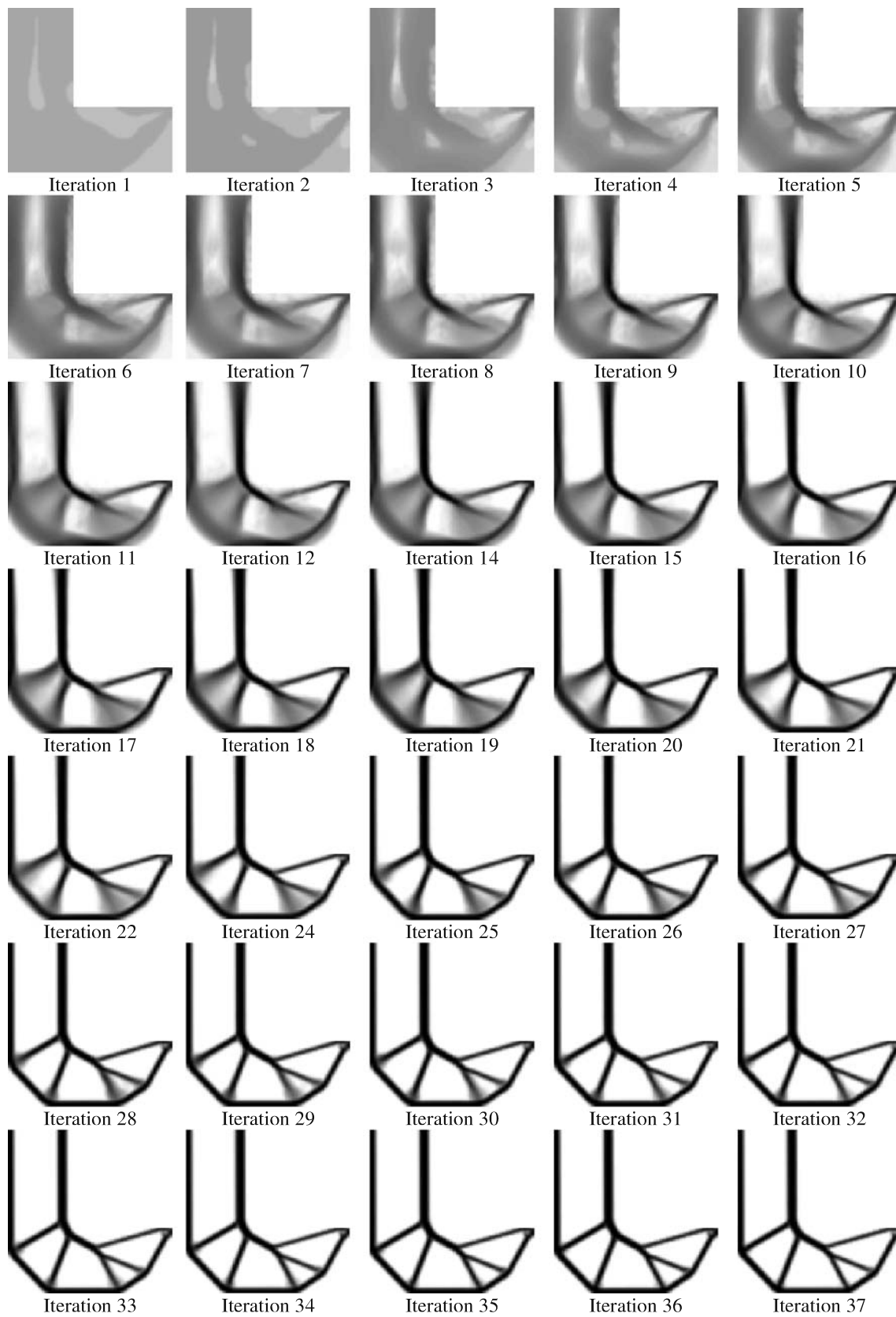
where  $\Omega_k$  is the set of elements in region  $k$  of our body region  $\Omega$ . Consequently, we impose the normalized  $p$ -norm constraint of (9) over each region  $k$ , viz.

$$\sigma_{\max_k} \approx c_k \sigma_{PN_k} \leq \bar{\sigma}, \quad k = 1, 2, \dots, m \quad (12)$$

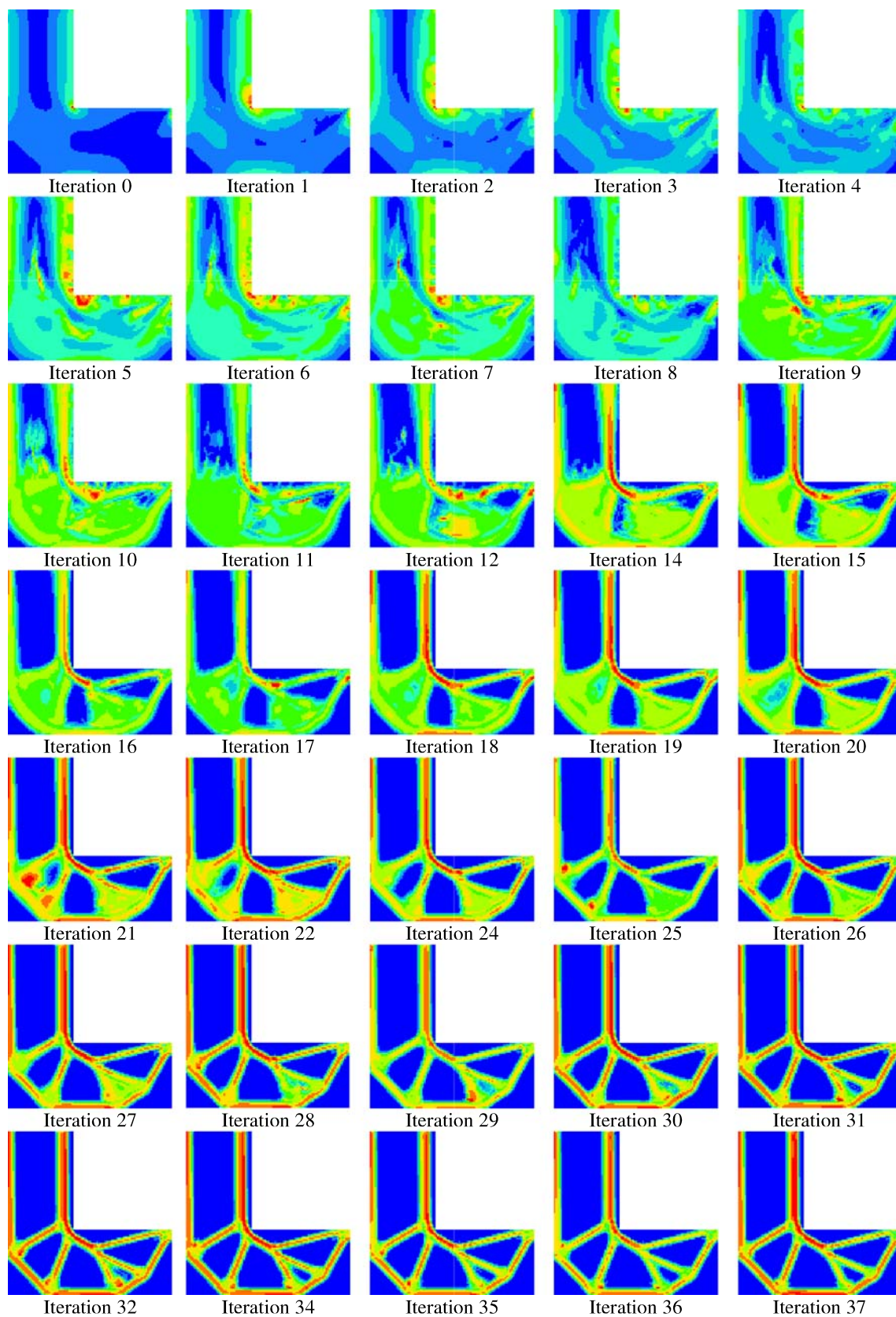
where  $c_k$  is calculated independently for each region using (10) and  $\sigma_{PN_k}$  is obtained from (8) by only considering the elements in region  $\Omega_k$ .

We implemented a number of region definitions based on physical location, stress distribution, and/or element connectivity, etc. and we found that a simple interlacing approach generated the best results. It should be noted that the goal of defining regions is to provide better control of local stress, thus regions need not be connected, i.e. the elements that make up each region need not be contiguous. To define the interlacing regions we sort elements based on their stress level at the current design iteration  $I$ :

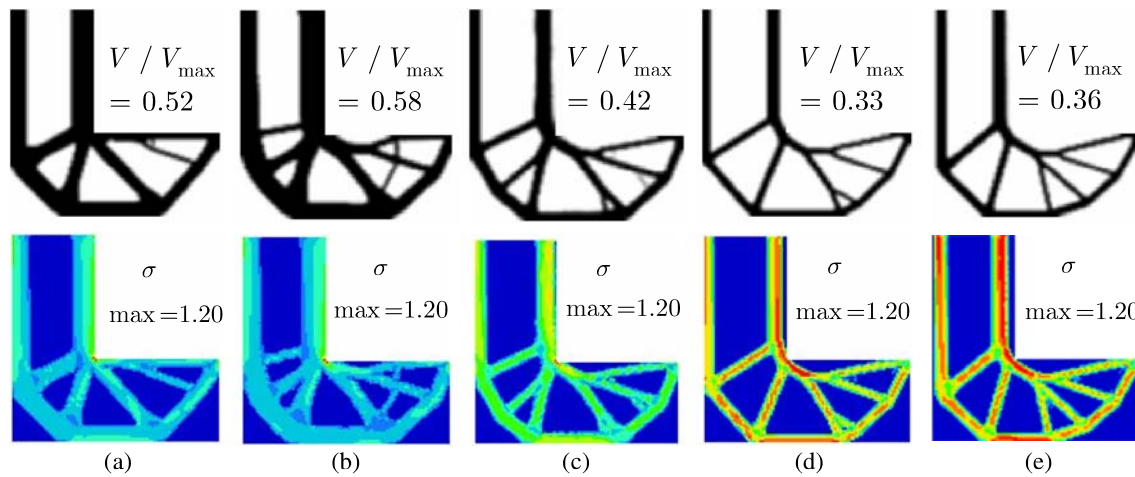
$$\{e_1, e_2, \dots, e_n : \sigma_{e_1}^I \leq \sigma_{e_2}^I \leq \dots \leq \sigma_{e_n}^I\} \quad (13)$$



**Fig. 5** Normalized global stress measure: density  $\rho$  evolution



**Fig. 6** Normalized global stress measure: stress  $\sigma$  evolution. (The color scales range from the minimum to the maximum stress for each iteration)



**Fig. 7** Density (top) and stress  $\sigma$  (bottom) for designs obtained with  $m = 1, 2, 4, 8,$  and  $16$  regions. **a**  $m = 1$  (295 iter.), **b**  $m = 2$  (108 iter.), **c**  $m = 4$  (105 iter.), **d**  $m = 8$  (110 iter.), **e**  $m = 16$  (108 iter.)

and then define the  $m$  regions as:

$$\Omega_k \equiv \{e_k, e_{m+k}, e_{2m+k}, \dots\}, k = 1, 2, \dots, m \quad (14)$$

For the values of  $m = 1$  and  $m = n$ , with  $n$  the number of elements, we revert to the constraints of (6) and (9). If the optimizer works perfectly, then as  $m$  increases the designs improve. However, in practice the optimizer may converge to a worse local minimum for a large  $m$  as shown in Fig. 7e. Also, as previously stated,  $m$  must not be too large lest the computational expense becomes intractable. We note that the regional constraint is similar to the “block aggregated” constraint in Paris et al. (2007), although our region definition differs. We also note that the proposed evolving region definition further adds to the potentially problematic non-differentiability of the constraints. However, we did not observe any adverse effects as the designs converge.

To illustrate the effect that the number of regions  $m$  has on the stress-constrained problem, we resolved the previous L-bracket example using the stress norm parameter  $P = 4$  with  $m = 1, 2, 4, 8,$  and  $16$  regions, cf. Fig. 7. We chose  $P = 4$  here to highlight the effect made by increasing the number of regions. The design obtained from  $m = 1$  region is unsatisfactory because a stress concentration exists at the reentrant corner that is not removed. As expected, the designs improve as the number of regions increases. Indeed, even for  $m = 8$  the stress concentration is effectively removed and the stress distribution becomes essentially uniform. In all designs, the maximum stress limit of 1.2 MPa is achieved (via the normalization). Thus, for a modest  $m$ -fold increase in our already efficient adjoint sensitivity computation we are able to significantly improve the

designs. Additionally, we are able to use a relatively modest stress norm parameter  $P = 4$  which results in a smooth design space which is easily traversed by the optimizer.

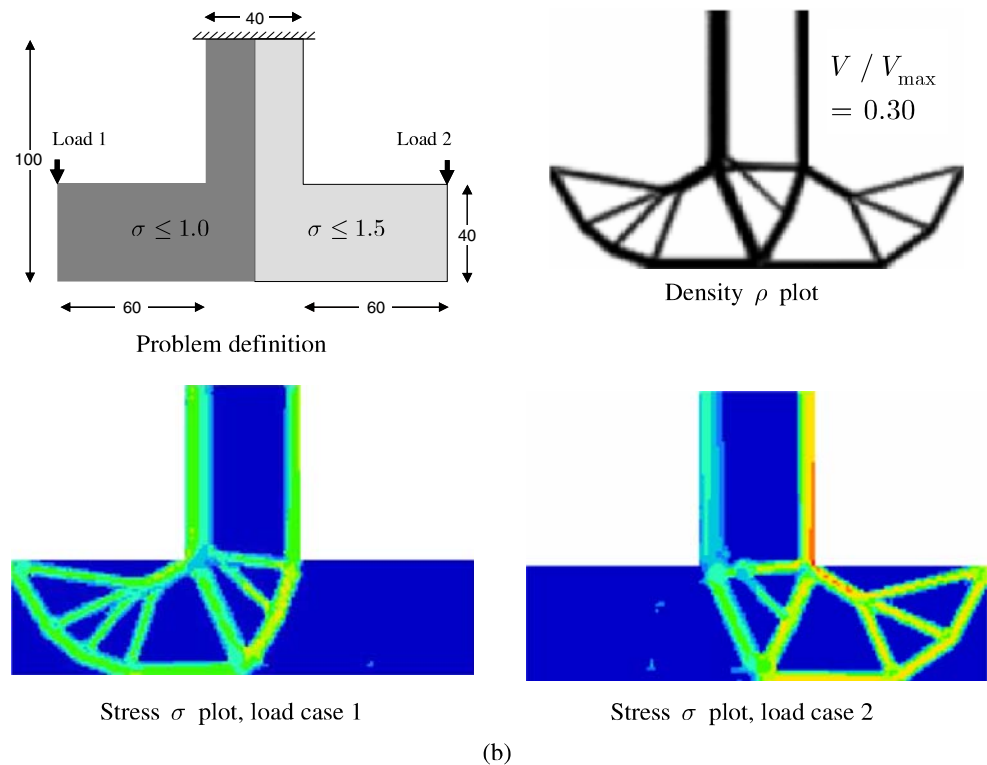
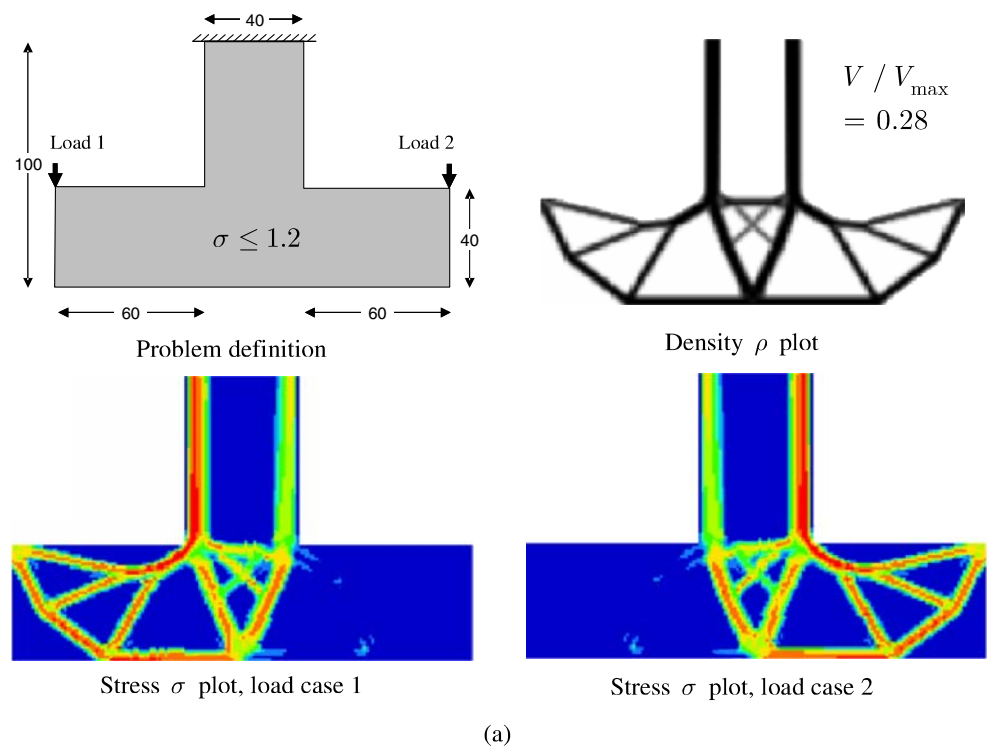
## 7 Multiple stress criteria and load cases

Our method is readily extended to accommodate multiple local constraint types and multiple load cases. In engineering design practice, there are scenarios in which one wishes to impose different stress constraints corresponding to different load cases, and possibly on different regions of the structure. For instance, one load case may correspond to a very severe load for which a stress constraint on the entire structure would be related to the material’s yield strength; additionally, there could be a load case related to a repetitive, damaging load for which a stress constraint on the regions where the structure would be welded to its support, would be related to the material’s endurance limit. To accommodate such instances, we could define our  $m$  regions  $\Omega_k$  accordingly. However, we have found that this practice produces regional biases that adversely affect the results. Hence, we redefine the original constraint of e.g. (6) as:

$$\frac{\sigma_e}{\bar{\sigma}_e} \leq 1, \quad e = 1, 2, \dots, n \quad (15)$$

Note that  $\sigma_e$  is the respective local response, e.g. it represents either the element von Mises stress or the strain energy density values, and  $\bar{\sigma}_e$  is its respective bound. To define the regions based on the interlacing scheme of (14), we now

**Fig. 8** Multiple stress criteria and multiple load cases. **a** Two load cases, single stress criterion ( $\bar{\sigma}_e = 1.2$  MPa), number of iterations = 190, **b** Two load cases, two stress criteria:  $\bar{\sigma}_e = 1.0$  MPa on the left;  $\bar{\sigma}_e = 1.5$  MPa on the right, number of iterations = 250



sort the elements by the value of  $\sigma_e / \bar{\sigma}_e$ , and redefine each regional constraint as:

$$c_k \left[ \sum_{e \in \Omega_k} v_{\sigma_e} \left( \frac{\sigma_e}{\bar{\sigma}_e} \right)^P \right]^{1/P} \leq 1, \quad k = 1, 2, \dots, m \quad (16)$$

cf. (8), (9) and (12). Here,  $\sigma_e$  is not required to be non-negative, but rather  $\sigma_e / \bar{\sigma}_e$  is assumed to be non-negative. We can also apply this technique in stress minimization problems wherein the  $\bar{\sigma}_e$  become scale factors and the cost function is defined from (16) with  $m = 1$  region.

A natural way to address the multiple load case problem is to consider each of the  $l$  load cases separately and use  $m$  regional constraints per load case. Alternatively, we could combine all load case  $l$  - element  $e$  stress ratios  $\sigma_e^l/\bar{\sigma}_e^l$  into a single set and subsequently divide it into “regions” via our interlacing strategy. In our example, the latter approach performs slightly better. Note that the minimum number of constraints in the former approach equals the number of load cases  $l$  whereas the minimum number of constraints in the latter approach equals 1. Hence the latter approach can be used to define the cost function for a stress minimization problem.

Figure 8 shows a modified L-bracket design that is identical to our previous examples with the obvious exceptions of its geometry and loading. Each of the two load cases is defined by the application of an equivalent concentrated load at their respective ends. To demonstrate the ability to apply different stress limits, the structure contains two regions divided by the vertical symmetry line. Two designs are generated: (1) volume minimization subject to stress constraints with a single stress criterion, i.e.  $\bar{\sigma}_e^l = \bar{\sigma} = 1.2$  MPa; and (2) volume minimization subject to stress constraints with elemental von Mises stress limits of  $\bar{\sigma}_e^l = 1.0$  MPa in the left region, and  $\bar{\sigma}_e^l = 1.5$  MPa in the right region. Designs are obtained using the stress norm parameter  $P = 8$  and  $m = 1$  regions (i.e. a single constraint for two load cases.) As seen in Fig. 8b, more material is distributed to the weaker left region to accommodate the lower stress limit. In both designs, the scaled stress, i.e.  $\sigma_e^l/\bar{\sigma}_e^l$ , tends to be uniform under the respective load. Convergence is obtained after 190 and 250 iterations; thus, we see that these multiple load case and multiple stress criterion problems are solved without appreciably more computational effort than compliance problems.

## 8 Examples

In this section we summarize our stress-constrained and stress minimization problems and present additional examples. For the stress-constrained problem we minimize volume subject to stress constraints:

$$\begin{aligned} \min_{0 < \mathbf{d} \leq 1} V(\boldsymbol{\rho}) \\ \text{s.t.} \\ c_k \sigma_{PN_k} = \left[ \sum_{(e,l) \in \Omega_k} v_e \left( \frac{\sigma_e^l}{\bar{\sigma}_e^l} \right)^P \right]^{1/P} \leq 1, \quad k = 1, 2, \dots, m \end{aligned} \quad (17)$$

where  $\boldsymbol{\rho}$  is the element density vector which is a function of the element design vector  $\mathbf{d}$ , cf. (1);  $V$  is the structure volume (which is a function of  $\boldsymbol{\rho}$  and hence  $\mathbf{d}$ );  $v_e$  is the element

$e$  solid volume;  $\sigma_e^l$  is the stress measure in element  $e$  under load case  $l$  (which is an implicit function of  $\boldsymbol{\rho}$  and hence  $\mathbf{d}$ , via the equilibrium equation);  $\Omega_k$  is the set of element - load case stresses in region  $k$ , which is defined according to Section 7;  $\bar{\sigma}_e^l$  is the stress limit for element  $e$  under load case  $l$ ;  $c_k$  is the normalization factor for region  $k$ , cf. (10); and  $m$  is the number of constraints, i.e. number of regions. Using the same notation we define the stress minimization problem as:

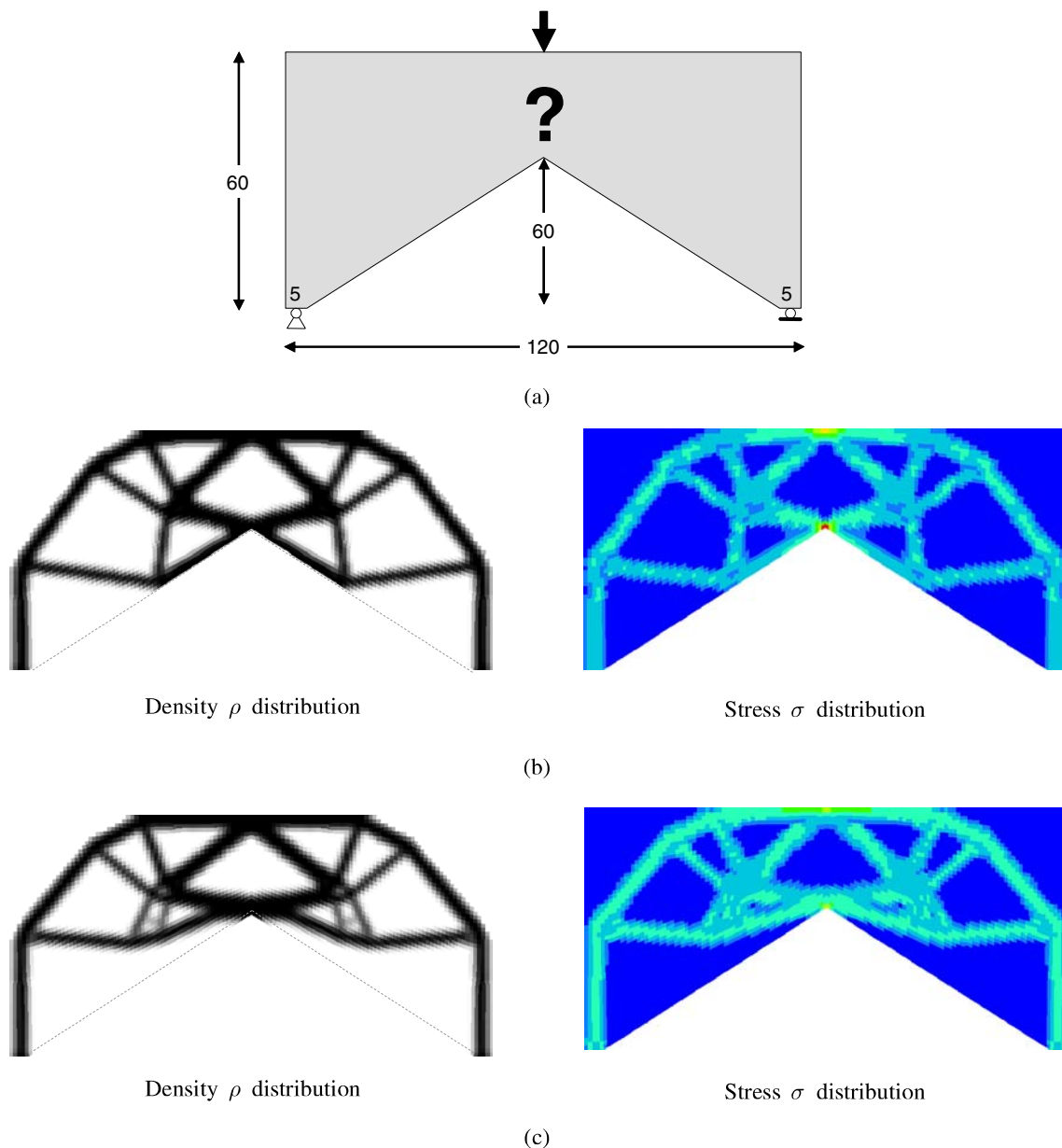
$$\begin{aligned} \min_{0 < \mathbf{d} \leq 1} \sigma_{PN} = \left[ \sum_{(e,l) \in \Omega} v_e \left( \frac{\sigma_e^l}{\bar{\sigma}_e^l} \right)^P \right]^{1/P} \\ \text{s.t.} \\ V(\boldsymbol{\rho}) \leq \bar{V} \leq V_{\max} \end{aligned} \quad (18)$$

where  $\bar{V}$  is the volume limit;  $V_{\max}$  is the design domain volume; and  $\bar{\sigma}_e^l$  is now interpreted as a scale factor. For completeness, we also define the compliance minimization problem which we use for comparison purposes:

$$\begin{aligned} \min_{0 < \mathbf{d} \leq 1} U = \int_{\Omega} \nabla u \cdot \mathbb{C}[\nabla u] du \\ \text{s.t.} \\ V \leq \bar{V} \leq V_{\max} \end{aligned} \quad (19)$$

where we recall that  $\mathbb{C} = \eta_c(\rho) \mathbb{C}_o$ . In the above (17), (18), (19), we also note that the elasticity equations define additional constraints that relate the stress values to the design  $\mathbf{d}$ . The sensitivities for these implicitly defined relations are obtained via the adjoint method (Michaleris et al. 1994).

We now apply the proposed procedure to design two additional structures. The first design concerns a portal frame and the second design concerns the well-known MBB beam. The portal frame design problem is depicted in Fig. 9a. The plane stress finite element analysis uses bilinear four-node quadrilateral elements with a thickness of 1.0 mm and an average edge length of 1.0 mm; the material model has a Young's modulus of  $E = 1.0$  MPa and a Poisson's ratio of  $\nu = 0.3$ ; the filter radius is  $r_f = 2.0$  mm; and the 8 N load is distributed symmetrically over nine nodes to avoid stress concentration. Two designs are generated: the first minimizes compliance subject to a volume constraint; and the second minimizes the maximum elemental von Mises stress, which is approximated using the stress norm parameter  $P = 8$ , subject to the same volume constraint. This example clearly illustrates the difference between minimum compliance and minimum stress designs. Not surprisingly, the stress design exhibits a higher compliance of 4,584 Nmm compared to the 4,154 Nmm of the compliance design, whereas the maximum stress of 3.32 MPa in the stress design is significantly lower than the 5.84 MPa in the compliance design. Qualitatively, we see that the stress design evenly distributes stress by avoiding stress concentrations

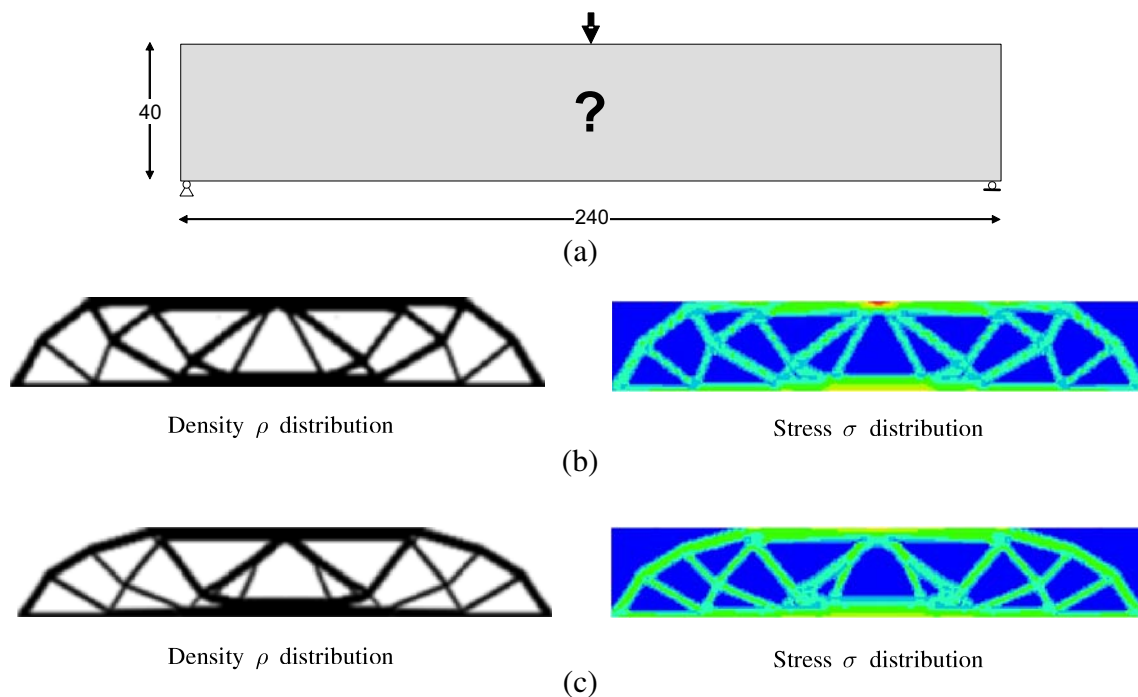


**Fig. 9** Portal frame: minimum compliance design vs. minimum stress design. **a** Problem definition, **b** minimum compliance design:  $V/V_{\max} = 0.3$ ; compliance = 4,154 Nmm; maximum stress = 5.84 MPa, **c** minimum stress design:  $V/V_{\max} = 0.3$ ; compliance = 4,584 Nmm; maximum stress = 3.32 MPa; number of iterations = 275

as opposed to the compliance design, which exhibits two stress concentration areas: one directly below the applied load and the other at the reentrant corner. In comparison to the compliance design, the minimum stress design avoids the stress concentrations by increasing the material adjacent to the load application area and by reconfiguring the two bottom chord members to form an arch over the reentrant corner.

The MBB beam design is depicted in Fig. 10. The plane stress finite element analysis uses bilinear 4-node square

elements with a thickness of 1.0 mm and edge length of 1.0 mm; the material model has a Young's modulus of  $E = 1.0$  MPa and a Poisson's ratio of  $\nu = 0.3$ ; the filter radius is  $r_f = 2.0$  mm; and the 8 N load is distributed symmetrically over nine nodes to avoid stress concentration. Again, two designs are generated: a compliance minimization subject to a volume constraint, and a volume minimization subject to 1.5 MPa maximum elemental von Mises stress constraint (which is enforced using the normalized constraint of (17) with  $P = 8$  and  $m = 1$ ). The stress design is performed



**Fig. 10** MBB beam: minimum compliance design vs. stress-constrained design. **a** Problem definition, **b** minimum compliance design:  $V/V_{\max} = 0.38$ ; compliance = 4,978 Nmm; maximum stress = 2.04 MPa, **c** Stress-constrained design:  $V/V_{\max} = 0.38$ ; compliance = 5,231 Nmm; maximum stress = 1.50 MPa; number of iterations = 160

first to obtain the minimized volume  $V = 0.38V_{\max}$  which in turn is used as the limiting volume  $\bar{V} = 0.38V_{\max}$  in the compliance design. Again, as expected, the stress design shows a more uniform stress distribution with a lower maximum stress, i.e. 1.50 MPa versus 2.04 MPa, and higher compliance, i.e. 5,231 Nmm versus 4,938 Nmm.

## 9 Conclusion

The paper describes a practical and computationally efficient procedure for solving stress-based topology optimization problems. The numerical examples demonstrate that the proposed method effectively removes the stress concentrations and generates black-and-white designs for practically sized problems. Our normalization and interlacing regional stress measure strategies effectively control the maximum stress. Our approach accommodates multiple load cases, different stress constraints in different regions of the design space, and multiple local criteria such as the von Mises stress, principal stress and strain energy density. The proposed method is versatile in that it can accommodate different filtering strategies, interpolation schemes for the stress relaxation, and aggregate functions for the stress.

**Acknowledgement** The authors would like to thank Caterpillar Inc. for their support of this work.

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